System Components & System Services

System Components & System Services

- Process Management
- Memory Management
- File Management
- I/O Management
- Storage Management
- Protection & Security
- Networking
- System Services

What Are OS Components?

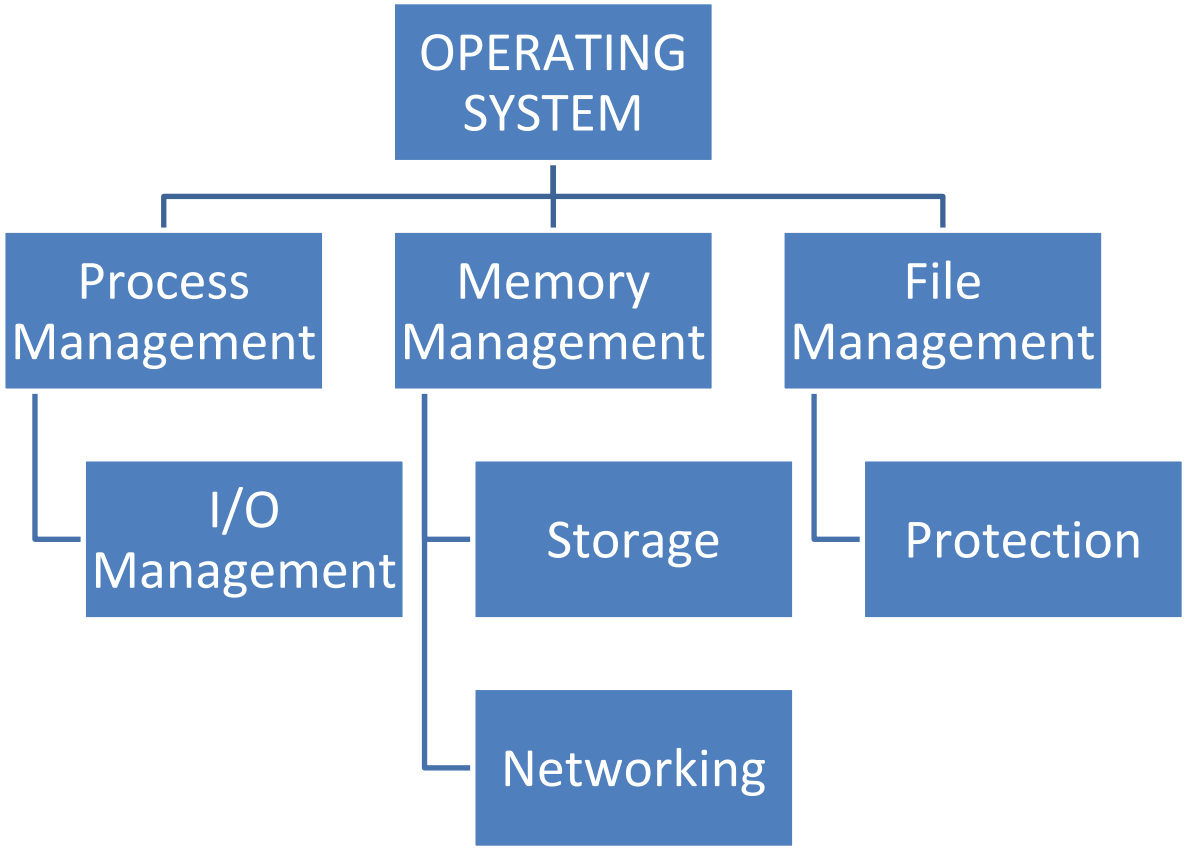
Definition:
Operating System components are the major subsystems responsible for managing different resources and providing different functions.

Major Components:

- Process Management
- Main Memory Management
- File Management
- I/O System Management
- Secondary Storage Management
- Protection & Security
- Networking

Think of an Operating System as a large organization.
It has different departments, and each department has a specific responsibility.
Similarly, an OS has different components responsible for different resources.

Operating System Components



Each component manages a particular **resource or responsibility**.

1 Process Management

What is a Process?

A **process is a program in execution.**

Process Management is responsible for:

- Creating processes
- Terminating processes
- Scheduling processes
- Suspending/resuming processes
- Process synchronization
- Inter-process communication

Example

When you open:

Chrome → Process

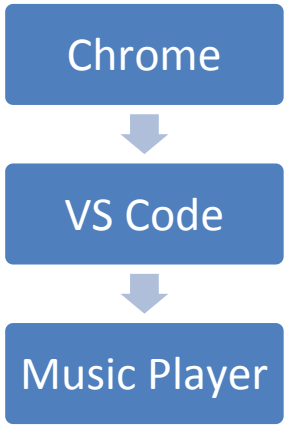
VS Code → Process

Music Player → Process

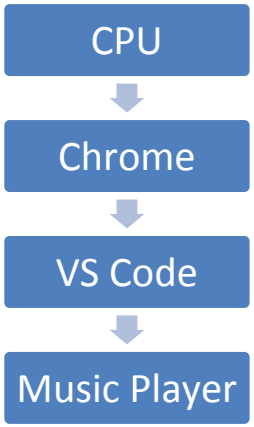
A program stored on your SSD is just a passive file.
When you execute it, it becomes a process.
The OS has to decide when and how that process gets CPU time.
This will also prepare students for your later **Process Management** unit

2 CPU and Process Management

Suppose three programs are running:



The OS decides:



This is done through **CPU scheduling**.

Question:

Who decides which process gets the CPU?

-> Operating System

3 Main Memory Management

Main Memory = RAM

The OS manages:

- Allocation of memory
- Deallocation of memory
- Tracking memory usage
- Protection of memory
- Sharing of memory

RAM

Operating System
Chrome
VS Code
Music Player
Free Space

If Chrome needs 2 GB of RAM, who gives that memory to Chrome?
The Operating System.

When Chrome closes, who makes that memory available again?
The Operating System.

4 File Management

What does the OS do with files?
The OS manages:

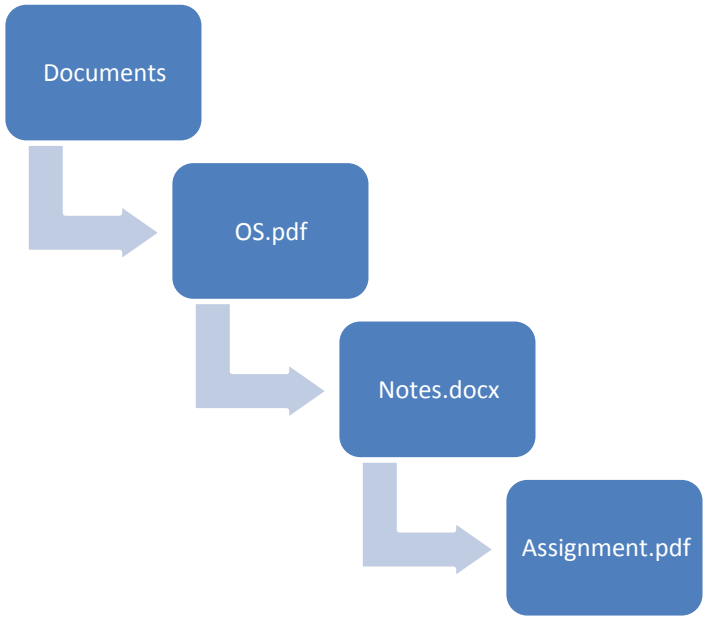
- File creation
- File deletion
- File reading
- File writing
- File organization
- Directory management
- File permissions

Example:
When you create:
OperatingSystem.pdf
the OS manages where and how that file is stored.

5 File System

File System
A file system organizes and manages data stored on storage devices.
Examples

- NTFS
- FAT32
- exFAT
- ext4
- APFS



Important
File System ≠ Operating System
The file system is one part of storage/file management.

6 I/O Device Management

I/O = Input / Output

Input Devices

- Keyboard
- Mouse
- Microphone
- Scanner

Output Devices

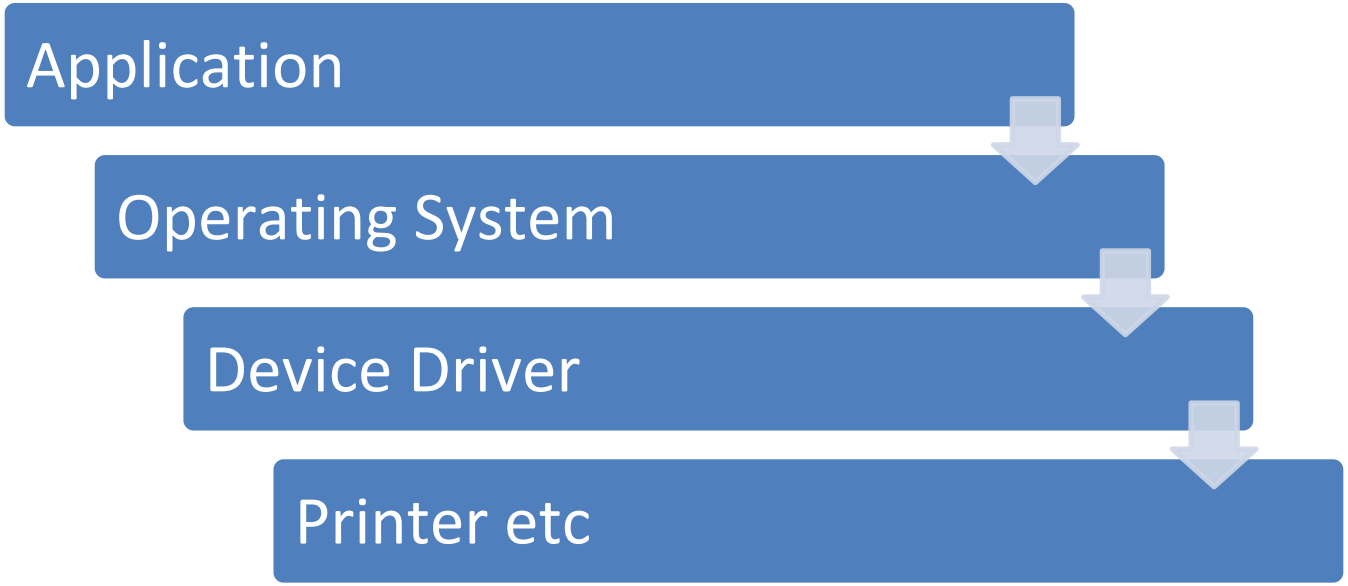
- Monitor
- Printer
- Speaker

OS Responsibilities

- Device communication
- Device drivers
- Buffering
- I/O scheduling
- Device allocation

7 Device Drivers

What is a Device Driver?
A **device driver is software that allows the Operating System to communicate with a hardware device.**



Why can’t an application directly control every hardware device?

8 Secondary Storage Management

Examples

- HDD
- SSD
- USB Drive
- Memory Card

OS Responsibilities

- Free-space management
- Storage allocation
- Disk scheduling
- Partition management

Example

When you save a 2 GB video, the OS must determine where the data will be stored.

9 Protection & Security

Protection

Controls **how processes/users access system resources.**

Security

Protects the system from:

- Unauthorized access
- Malware
- Data theft
- Attacks

Examples

- Password
- File permissions
- User accounts
- Access control
- Encryption

Important Difference

Protection: Controls access to resources.

Security: Protects the system and data against threats.

10 Networking

Modern operating systems provide networking capabilities.

Functions

- Network communication
- Resource sharing
- Remote access
- Network configuration
- Communication between systems

Example

When you open:
google.com
your OS participates in the network communication required by the application.

Summary of System Components

Component	Main Responsibility
Process Management	CPU/processes
Memory Management	RAM
File Management	Files/directories
I/O Management	Devices
Storage Management	Secondary storage
Protection & Security	Access/control
Networking	Communication

What Are System Services?

Definition:
System services are facilities provided by the Operating System to users and application programs.
They make it easier for programs and users to perform common operations.

Major System Services:

1. User Interface
2. Program Execution
3. I/O Operations
4. File-System Manipulation
5. Communication
6. Error Detection
7. Resource Allocation
8. Accounting
9. Protection & Security

1 User Interface

The OS provides an interface through which users interact with the system.

Types:

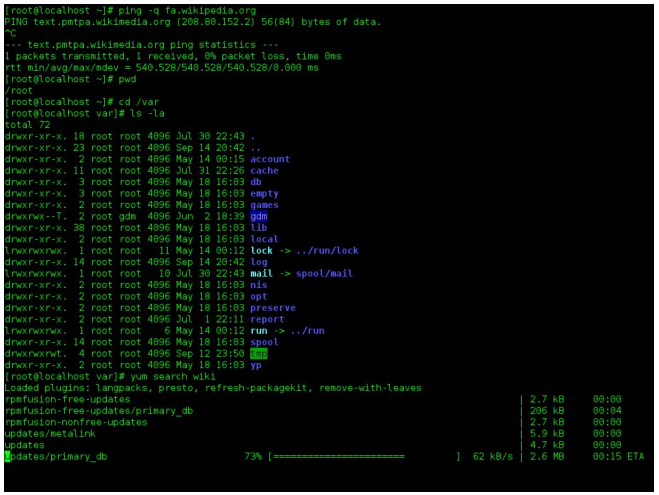
CLI – Command Line Interface

Example:

ls

cd Documents

pwd



GUI – Graphical User Interface

Examples:

- Windows Desktop
- macOS Finder
- Linux Desktop

Touch Interface

Examples:

- Android
- iOS



2 Program Execution

The OS provides services to:

- Load a program into memory
- Start execution
- Manage execution
- Terminate the program

Example

Double-click Chrome

↓

OS loads program

↓

Program starts

↓

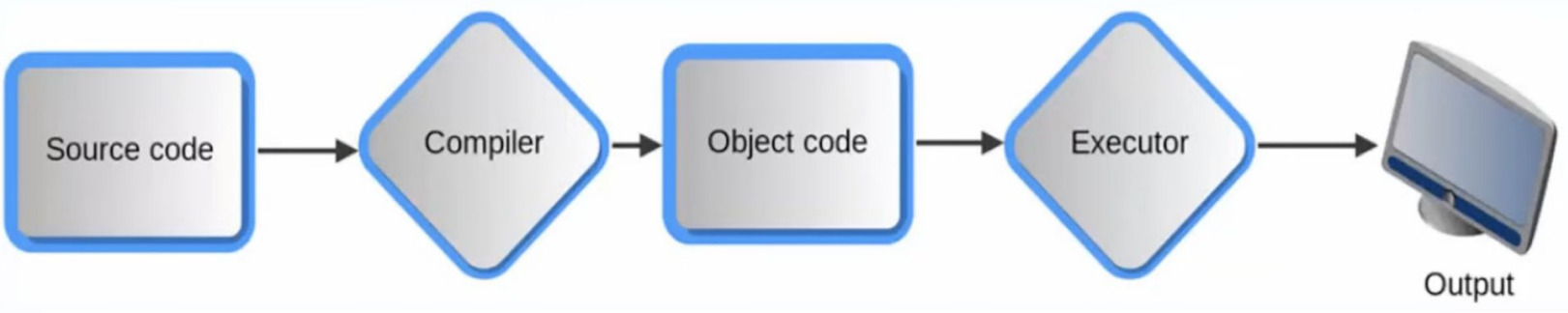
Chrome runs

↓

User closes Chrome

↓

OS terminates process

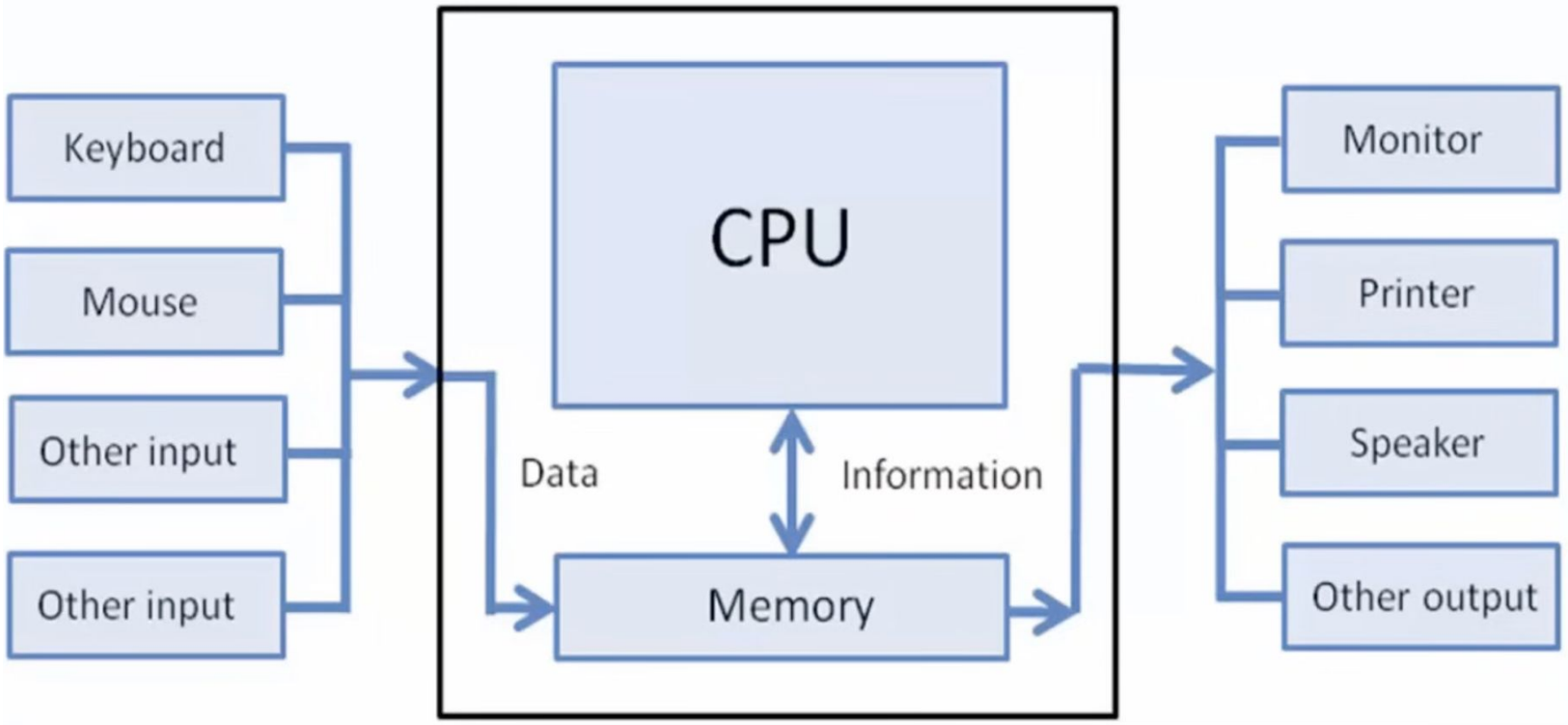

Source code
Compiler
Object code
Executor
Output

3 I/O Operations

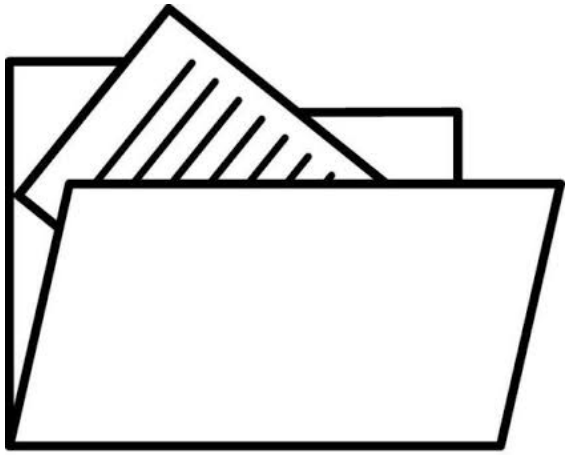
Programs frequently need I/O operations.
Examples:

- Read keyboard input
- Display output
- Read a file
- Write to disk
- Print a document

Important
The application does not need to understand every hardware detail.
The OS provides a standard interface for I/O operations.


Keyboard
Mouse
Other input
Other input
CPU
Data
Information
Memory
Monitor
Printer
Speaker
Other output

4 File-System Manipulation



The OS provides services to:

- Create files
- Delete files
- Open files
- Read files
- Write files
- Create directories
- Delete directories
- Rename files

Example

When you click:

Save → lecture1.pptx

multiple OS services are involved.



5 Communication

Processes may need to communicate with:

- Other processes on the same computer
- Processes on another computer

Two Major Mechanisms

IPC – Inter-Process Communication

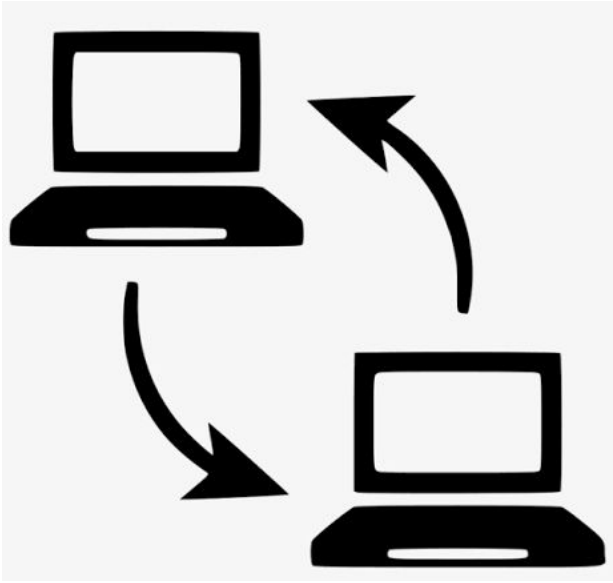
Examples:

- Shared Memory
- Message Passing

Network Communication

Examples:

- TCP/IP
- Sockets



6 Error Detection



The OS constantly monitors for errors.

Examples

- Memory errors
- CPU errors
- Disk errors
- I/O errors
- Network errors
- Program errors

Example

If you try to access an invalid memory location, the OS can detect the violation and terminate or otherwise handle the faulty process.

7 Resource Allocation

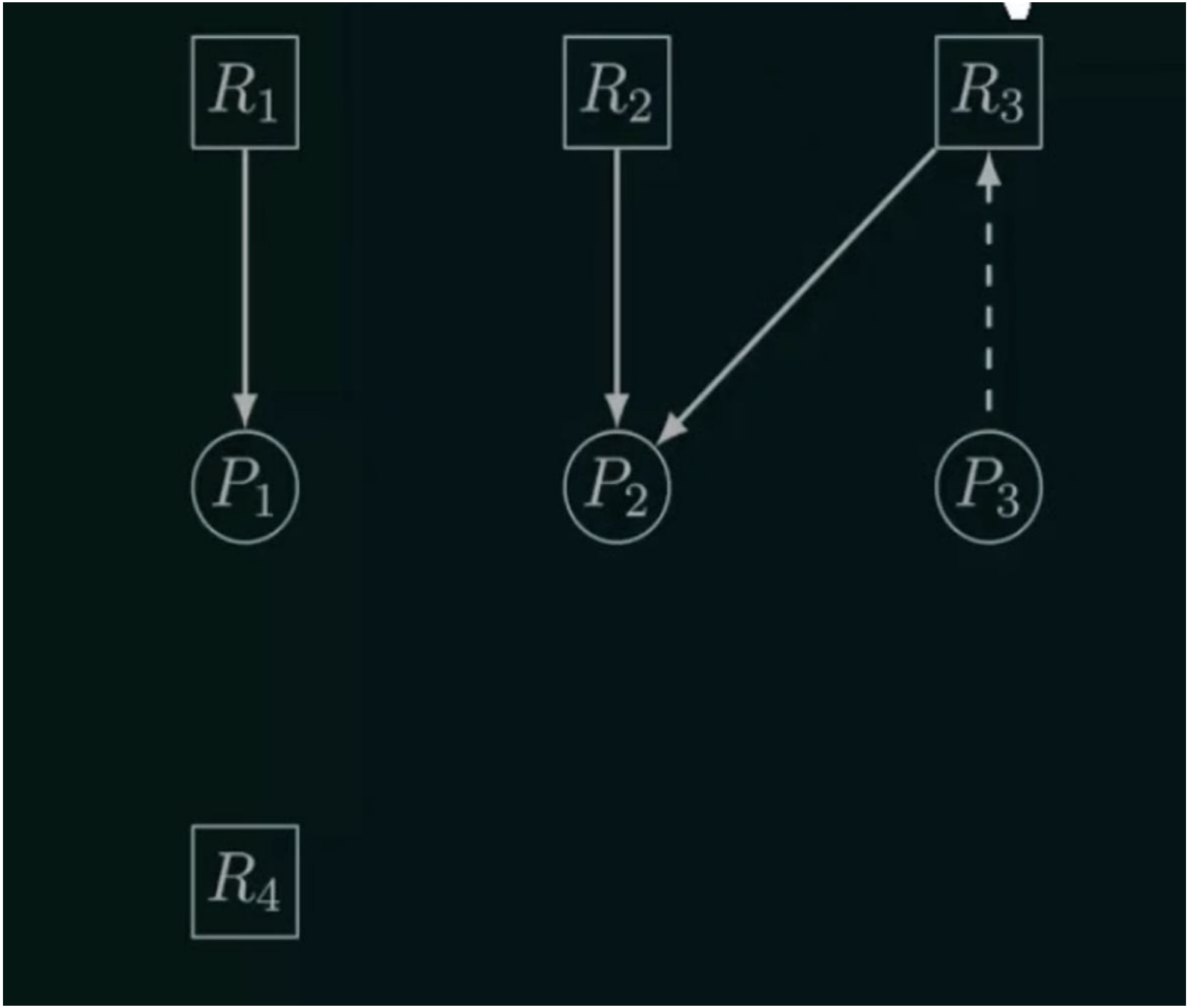
Many programs compete for limited resources.
Resources:

- CPU
- RAM
- Storage
- I/O devices

Example:

CPU
↙ ↓ ↘
Chrome VS Code Music

The OS decides how resources are allocated.
Key Idea
OS = Resource Manager


R_1
R_2
R_3
P_1
P_2
P_3
R_4

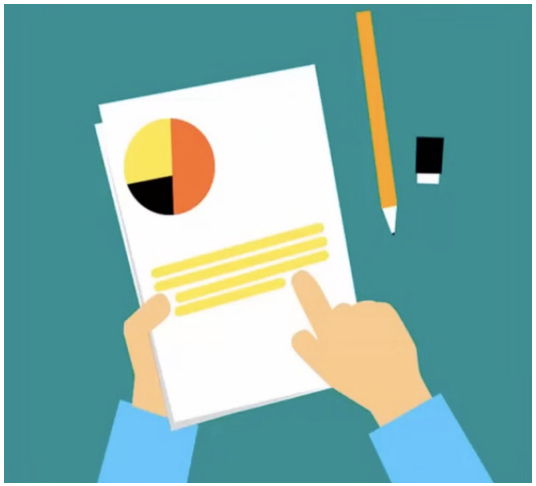
8 Accounting

What is Accounting?

Keeping track of how system resources are used.

Examples

- CPU usage
- Memory usage
- Disk usage
- Network usage
- User activity



Example:

Task Manager shows:

CPU 32%
Memory 68%
Disk 5%
Network 2 Mbps

9 Protection

The OS ensures that users/processes access only the resources they are permitted to access.

Example:

User A

❌ Cannot access User B’s private files.



Protection Mechanisms:

- Authentication
- Authorization
- File permissions
- Access control

Components vs Services

Components:
How the OS is organized internally.
Examples:

- Process Management
- Memory Management
- File Management

Services:
What the OS provides to users/programs.
Examples:

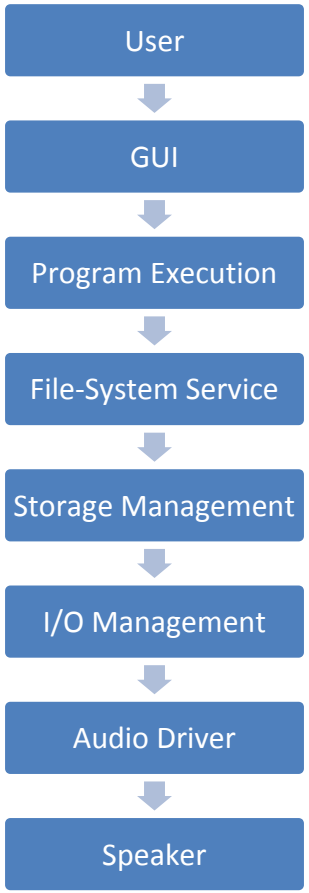
- Program execution
- I/O operations
- File manipulation
- Communication

Remember:
Components = What the OS manages
Services = What the OS provides

Real-Life Example

Opening a Music File
Suppose you double-click:
Song.mp3
What happens?

How many OS components/services are involved in this simple action?



Q1. Which OS component manages processes?

A. File Management. B. Process Management. C. Storage Management D. Networking

Q2. Which component manages RAM?

A. Memory Management B. File Management. C. I/O Management. D. Security

Q3. Which of the following is an OS service?

A. Program Execution. B. CPU Core. C. RAM. D. SSD

Q4. Which software allows the OS to communicate with hardware?

A. Compiler. B. Device Driver. C. Browser. D. Shell Script

Q5. Which service allows processes to exchange information?

A. Accounting. B. Error Detection. C. Communication. D. Booting

B

A

A

B

C

Summary

System Components

- Process Management
- Memory Management
- File Management
- I/O Management
- Secondary Storage Management
- Protection & Security
- Networking

System Services

- User Interface
- Program Execution
- I/O Operations
- File-System Manipulation
- Communication
- Error Detection
- Resource Allocation
- Accounting
- Protection